Zhanwei Zhang (张展玮)

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Education

M.Phil. in Data Science and Analytics, Hong Kong University of Science and Technology (Guangzhou) **Sep 2025 ~ Present**

Information Hub

Advisor: Prof. Zishuo Ding

B.Sc. in Computer Science and Technology, Southern University of Science and Technology (SUSTech) **Sep 2021 ~ Jun 2025**

Turing Master Class

Advisor: Prof. Yepang Liu

GPA: 3.79 / 4.0 | Weight Avg Score: 90.92 | Ranking: 36 / 195

Selected Coursework: Data Structures & Algorithms (A), Machine Learning (A), Database Systems (A-), Compilers (B+)

Experience

Qualcomm, Shenzhen | AI Agent Platform Intern **May 2026 ~ Present**

Led architecture evolution of an internal AI Agent platform, enabling unified Agent / Workflow / Tool integration.

Designed an Engine + Plugin architecture with dynamic backend/frontend loading via Python Entry Points and Module Federation.

Built an Agent Runtime with DeepAgents, unifying Tool / MCP / Skills / Subagent integration with AG-UI and HITL support.

Lingsome, Shenzhen | LLM Intern **Aug 2024 ~ Aug 2025**

Designed multi-granularity RAG indexing and joint retrieval strategies to improve retrieval quality on domain data.

Researched and adapted Microsoft GraphRAG to company-specific domain data, constructing domain knowledge graphs.

Explored automated RAG evaluation through question generation and Ground Truth annotation prototypes.

Wuhan University, Wuhan | Visiting Researcher **May ~ Aug 2024**

Advisor: Prof. Jinfu Chen (WHU); Prof. Weiyi Shang (UWaterloo)

Studied software logging and failure-workaround patterns, building an automated pipeline to mine and sample relevant code commits.

Selected Projects

OpenStar **Jul 2026 ~ Present**

Built Chat / Explore / Watch Agent workflows for open-source project discovery, understanding, and tracking.

Designed Agent research and Watch pipelines that autonomously inspect releases and repository activity via tools and produce structured assessments.

NIYO.AI Custom-Fit Nail 3D Modeling Platform **Jan 2026 ~ Present**

Turned vision and geometry prototypes into a production 3D annotation/modeling platform, covering full-stack development and the vision-to-3D workflow.

Built synthetic-data and training pipelines for millions of multi-view samples, model training/evaluation, and production inference.

Rewrote the shared geometry core in Rust with Python/WASM bindings, cutting compute from hundreds of ms to ~10 ms for real-time 3D interaction.

WiTH - AI Health Companion **Jun 2026**

Built an AI health Agent integrating user profiles, health records, and 20+ tools for personalized interaction.

Implemented semantic memory and proactive reminders that turn conversations into actionable tasks; won Third Prize at the Shenzhen Hackathon.

AI Micro-Drama Studio **Sep 2025 ~ Jan 2026**

Built an end-to-end pipeline turning novels/scripts into storyboards, integrating generation, editing, and export in one workflow.

Othello Game through Java and Python Programming with Strong AI **Mar 2023**

Implemented Monte Carlo and Alpha-Beta search; ranked 3/29 with an 81% win rate.

Canteen Traffic Monitoring **Dec 2023 ~ Jan 2024**

Calculated the length of the queue by monitoring data and displayed a chart showing the changes in queue length.

Won award for finalist in National College Students' Innovation and Entrepreneurship Training program.

About 30,000 visits within three months.

Research

Numerical Error Detection in Floating-Point Computing **Sep 2024 ~ Feb 2026**

Developed perturbation- and Newton-guided error detection methods, achieving 173/174 significant-error cases at 0.13% of oracle cost and detecting 80 bugs across 47 functions.

R1-style Reasoning Pipeline Reproduction **May ~ Jun 2025**

Implemented Cold-start SFT and GRPO on Qwen2.5 models (0.5B–7B) to reproduce and evaluate an R1-style reasoning pipeline.

LLM-Based JSON Parser Fuzzing **Sep 2023 ~ Jan 2024**

Used Llama 2 7B/13B to fuzz 13 JSON parsers, testing 100+ case types and identifying 26+ behavioral differences.

Selected Publications

Y. Tan, **Z. Zhang**, et al. *"ICE: Reducing Search Space for Error-Inducing Input Detection."* IEEE Transactions on Software Engineering, 2026.

Y. Tan, **Z. Zhang**, et al. *"Mathematically-Guided Detection of Floating-Point Errors."* ISSTA 2026.

Patents

一种点餐方法、系统、终端及介质 (Innovative Ordering Method, System, Terminal, and Medium Patent) **May 2023**

Innovated a method and system to alleviate peak-hour traffic in cafeterias.

Applied on May 5, 2023; Application no: 202310498065

Skills

Programming Languages: Python, TypeScript/JavaScript, Rust, C/C++, Java

Technology Stack: React / Next.js, PyTorch, PostgreSQL, Docker, Git

AI Development: Codex, Claude Code, OpenAI / Claude Agent SDK, Pi Agent, LangChain

Languages: Mandarin (Native), Cantonese (Native), English (IELTS 6.5)

Honors & Scholarships

Third Prize, Shenzhen Hackathon **Jun 2026**

Special Innovation Award (Unique Winner) & Second Prize, OpenHarmony Competition Training Camp **Sep 2025**

Postgraduate Studentship (PGS), HKUST(GZ) **Sep 2025**

Outstanding Student, SUSTech **Jan 2024**

Honorable Mention, Mathematical Contest in Modeling **May 2023**

Finalist, National College Students' Innovation and Entrepreneurship Training program **Jun 2023**

Third Prize, China Undergraduate Mathematical Contest in Model **Sep 2023**